

RAGR: Review-Augmented Generative Recommendation

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Sequential recommendation (SR) is traditionally formulated as next-item prediction over chronological item interactions. Although recent generative recommendation (GR) methods introduce new machinery, such as semantic IDs, autoregressive decoding, and unified token spaces, they largely inherit the same item-only modeling assumption. We argue that this design constitutes a structural bottleneck, because user decision-making is not purely behavioral: while item interactions reveal what users choose, review feedback often explains why they choose it by exposing latent evaluative factors.

Motivated by this observation, we propose *Review-Augmented Generative Recommendation (RAGR)*, a novel GR framework that incorporates review feedback into the generative user sequence rather than treating reviews as auxiliary side information. Specifically, RAGR introduces a *Review-Augmented User Sequence Modeling* mechanism that interleaves item semantic IDs and review semantic IDs in chronological order to construct a mixed behavioral-semantic sequence, enabling review signals to participate directly in autoregressive next-token generation. To preserve the recommendation objective, we further introduce an *Item-Centric Task Generation Alignment* strategy based on direct preference optimization (DPO), encouraging the model to favor item tokens over review tokens at prediction positions. Experiments on three real-world datasets show that RAGR yields consistent and significant gains over strong GR backbones. Our code is available at <https://github.com/Zhang-Yingyi/RAGR>.

CCS Concepts: • **Information systems** → **Recommender systems**; **Personalization**.

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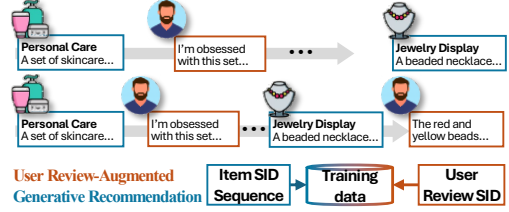
1 Introduction

Recommender systems are indispensable to modern e-commerce platforms [46], such as Amazon [10] and Alibaba [35], where they surface relevant items from catalogs of millions. Among various paradigms, sequential recommendation (SR) [16, 42] stands out because user preferences are inherently *dynamic*, shaped not only by static profiles but by the evolving temporal patterns of historical interactions [48, 49]. SR is predominantly cast as a *next-item prediction* task—given a chronological sequence of past interactions as shown in Fig. 1 a), the model forecasts the item a user will engage with next [6, 25]. This formulation is tightly coupled with the representation mechanism of conventional models: architectures such as SASRec [16] and BERT4Rec [32] rely on an *item ID embedding table*, under which a user’s history is encoded as an ID sequence and the learning objective reduces to predicting the next item ID, as illustrated in Fig. 1 c).

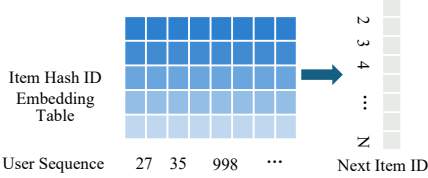
a) Existing Generative Recommendation



b) Ours Review-Augmented Generative Recommendation

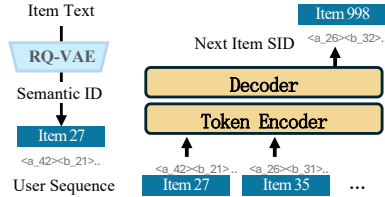


c) Traditional Sequential Recommendation



Limitation of ID-based SR: Item ID embedding tables capture behavioral outcomes, but cannot naturally encode free-form user review feedback.

d) Generative Sequential Recommendation



Gap in SID-based GR: Existing GR enhances item representation with SIDs, but still models users as item-only sequences, leaving review feedback—and the preference rationale it reveals—underutilized.

Fig. 1. Comparison between existing and our proposed generative recommendation paradigms. (a) Existing generative recommendation based on item-only sequences. (b) The overall framework of the proposed Review-Augmented Generative Recommendation (RAGR). (c) The proposed review-augmented user sequence modeling mechanism. (d) The item-centric task generation alignment via DPO.

With the emergence of generative recommendation (GR) [18, 29], this paradigm has been reparameterized through semantic ID (SID) [26, 29, 36], autoregressive decoding, and unified token spaces. As shown in Fig. 1 d), GR replaces the conventional item-ID lookup table with semantic tokenization and generation. However, despite this representational shift, the underlying modeling paradigm remains essentially the same as that in Fig. 1 a): existing GR methods still model the user history as an item-only trajectory and optimize for generating the next item. We argue that this inherited assumption is not merely a missed opportunity but a *structural bottleneck*. In reality,

user decision-making is inherently multi-faceted: purchases are often preceded by browsing and comparison, and followed by explicit feedback such as textual reviews [8, 47]. As McAuley and Leskovec [23] show, reviews expose *hidden evaluative dimensions*—such as quality, usability, and aesthetics—that shape a user’s rating but remain invisible in interaction logs alone. An item-only sequence captures only the *outcome* of a decision—*what* the user chose—while omitting the explanatory signals that reveal *why* it was chosen. Without such signals, the model must infer preference dynamics solely from item co-occurrence patterns, resulting in a shallow and brittle understanding of user intent.

Notably, the generative framework already possesses the machinery needed to transcend this bottleneck. A distinguishing property of GR is its *unified tokenizer*, which projects item text content into a shared token space [4, 18, 34]. This unified tokenization mechanism makes GR fundamentally different from conventional ID-based recommendation, because it allows recommendation objects to be represented and generated within a common semantic space. In current practice, however, this tokenizer is applied almost exclusively to items—mapping each item to SID [26, 29, 36] for autoregressive modeling—while the same mechanism could equally operate on other textual signals that users routinely produce, most notably post-interaction reviews. This is particularly appealing because reviews provide fine-grained preference signals beyond item transitions. Put differently, existing research has substantially advanced the *expressiveness of item representation* yet left the *expressiveness of the user sequence* largely untouched. This asymmetry gives rise to our central research question: ***Can the unified tokenizer of GR be repurposed to encode user review feedback as semantic tokens and weave them into the interaction sequence, thereby enabling the model to capture not only behavioral outcomes but also the underlying preference rationale?*** Addressing this question entails two intertwined challenges: (1) *Heterogeneous sequence construction*—items and reviews differ in granularity and functional role; projecting both through a shared tokenizer into a coherent sequence requires careful design to avoid introducing noise. (2) *Recommendation-objective preservation*—once review tokens enter the generative sequence, a principled alignment mechanism is needed to ensure that they serve as supporting evidence for next-item prediction rather than a competing generation target.

To address these challenges, we propose **Review-Augmented Generative Recommendation (RAGR)**, as illustrated in Fig. 1b. The central idea of RAGR is to move beyond the conventional treatment of reviews as auxiliary features in discriminative recommenders. Instead, we bring review feedback directly into the GR paradigm by encoding reviews as tokens within the user sequence itself—extending the item-only behavioral trajectory into a review-augmented one. Specifically, RAGR consists of two complementary components. **(i) Review-Augmented User Sequence Modeling.** Item SIDs and review SIDs are interleaved chronologically to form a mixed-signal user sequence, over which the GR backbone is trained to predict the next SID—allowing review context to directly participate in the autoregressive generation process. **(ii) Item-Centric Task Generation Alignment.** Because the mixed sequence may cause the model to devote generation capacity to reviews rather than items, we apply a direct preference optimization (DPO) [27] alignment mechanism that teaches the model to *prefer* generating item SIDs over review SIDs at prediction positions, ensuring that the learning objective remains anchored to next-item recommendation.

Our main contributions are summarized as follows:

- We identify a structural limitation of existing GR methods—their confinement to item-only sequences—and propose RAGR, to our knowledge the first framework to elevate reviews from auxiliary features in discriminative recommenders to tokens within the GR paradigm, extending user sequences from purely behavioral to mixed behavioral-semantic.

- We design a *Review-Augmented User Sequence Modeling* mechanism that interleaves item and review SIDs chronologically, enabling review context to participate directly in the autoregressive generation process and enrich the preference signal available at each prediction step.
- We propose an *Item-Centric Task Generation Alignment* strategy based on DPO, which steers the model to prefer item SIDs over review SIDs at prediction positions, ensuring that reviews inform—but never displace—the next-item recommendation objective.
- We conduct extensive experiments on three real-world datasets, demonstrating that RAGR yields consistent and significant gains when applied to multiple GR backbones. Ablation studies further confirm that review augmentation and task alignment are both indispensable.

The remainder of this paper is structured as follows. Section 2 defines the review-augmented generative recommendation problem. Section 3 introduces the proposed approach, which is evaluated in Section 4. Then, Section 5 summarizes the recent development of sequential recommendation and generative recommendation. Finally, conclusions are drawn in Section 6.

2 Problem Formulation

Let $\mathcal{U}$ and $\mathcal{I}$ denote the user set and item set, respectively. For each user $u \in \mathcal{U}$, we observe a chronological interaction history

$$\mathcal{S}_u = [(i_1, r_1), (i_2, r_2), \dots, (i_T, r_T)], \quad (1)$$

where $i_t \in \mathcal{I}$ denotes the interacted item at step t , and r_t denotes the associated textual feedback, such as a review. Different from conventional sequential recommendation, which models only the item sequence $(i_1, i_2, \dots, i_T)$, we consider a review-augmented user sequence in which item interactions and review feedback are jointly modeled. Given the historical sequence up to step $t-1$,

$$\mathcal{S}_u^{<t} = [(i_1, r_1), (i_2, r_2), \dots, (i_{t-1}, r_{t-1})], \quad (2)$$

our goal is to predict the next target item i_t . Formally, we aim to learn a generative recommendation model

$$f_\theta : \mathcal{S}_u^{<t} \mapsto i_t, \quad (3)$$

where θ denotes model parameters.

To support generative modeling, we represent both items and reviews in a unified token space. Let $\mathbf{z}(i_t)$ and $\mathbf{z}(r_t)$ denote the tokenized representations of item i_t and review r_t , respectively. Then the historical sequence can be rewritten as

$$\tilde{\mathcal{S}}_u^{<t} = [\mathbf{z}(i_1), \mathbf{z}(r_1), \mathbf{z}(i_2), \mathbf{z}(r_2), \dots, \mathbf{z}(i_{t-1}), \mathbf{z}(r_{t-1})]. \quad (4)$$

Accordingly, the recommendation objective is to generate the tokenized representation of the next target item:

$$p_\theta(\mathbf{z}(i_t) \mid \tilde{\mathcal{S}}_u^{<t}). \quad (5)$$

Based on this formulation, we construct the training set as

$$\mathcal{D} = \left\{ (\tilde{\mathcal{S}}_u^{<t}, \mathbf{z}(i_t), \mathbf{z}(r_t)) \mid u \in \mathcal{U}, 2 \leq t \leq T \right\}, \quad (6)$$

where $\mathbf{z}(i_t)$ is the tokenized target item and $\mathbf{z}(r_t)$ is the corresponding review of the user. The learning problem in this work is therefore to perform next-item generation over review-augmented user sequences while preserving the item-centric recommendation objective. ***However, incorporating review tokens into the unified generative space introduces ambiguity in the prediction target, since both item sequences and review sequences become plausible continuations under the same context.***

To resolve this ambiguity, the proposed RAGR framework addresses two sub-problems in sequence: (1) how to train a generative model over the mixed item–review SID sequence so that

review context enriches preference modeling (Section 3.3), and (2) how to align the generation objective so that the model favors item SIDs over review SIDs at prediction positions (Section 3.4).

3 The proposed RAGR method

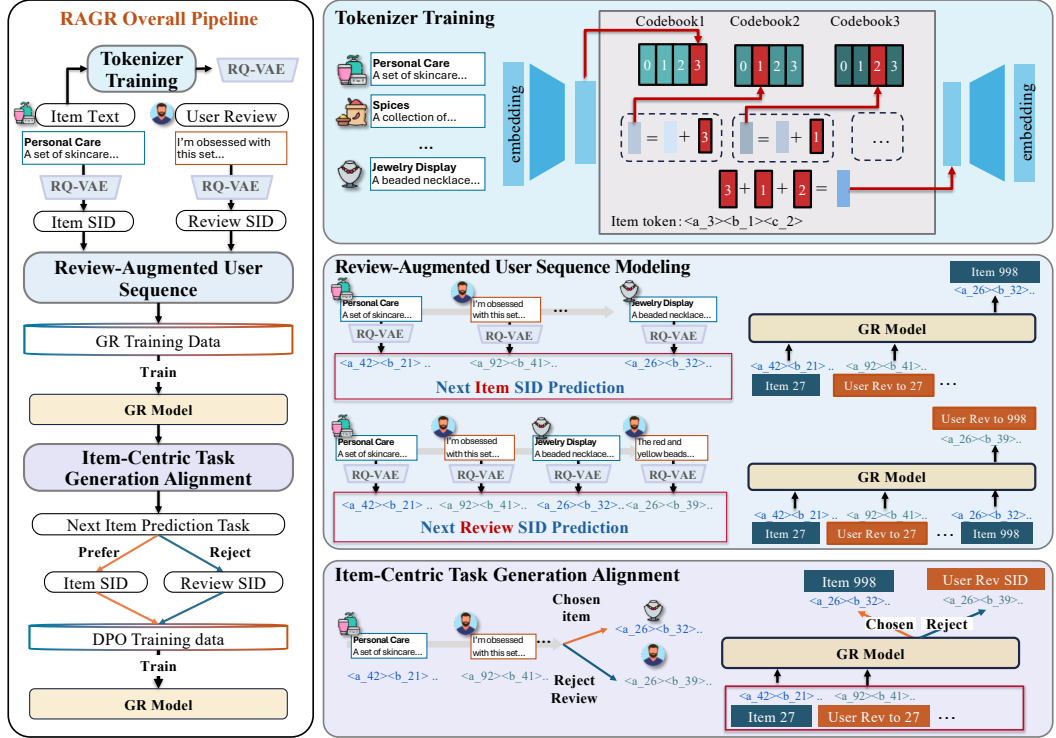


Fig. 2. The overall framework of the proposed **RAGR**, which consists of three main stages: Tokenizer Training, Review-Augmented User Sequence Modeling, and Item-Centric Task Generation Alignment.

3.1 Overview of RAGR

As illustrated in Fig. 2, RAGR consists of three stages: *Tokenizer Training*, *Review-Augmented User Sequence Modeling*, and *Item-Centric Task Generation Alignment*.

In the first stage, we **train a tokenizer** based on RQ-VAE [29] to map both items and reviews into a unified semantic ID space. This stage produces discrete token representations required for subsequent generative modeling. In the second stage, we construct **review-augmented user sequences** by interleaving item interactions with their associated review feedback. Based on such sequences, the generative recommender is trained for next item generation, so that recommendation is conditioned not only on item interactions but also on semantic evidence derived from review feedback. In the third stage, we introduce **item-centric task generation alignment** to preserve the task boundary of next item recommendation. Specifically, we employ DPO-based preference alignment to bias generation toward target items rather than review text, so that review feedback serves as evidence for item selection instead of becoming a competing generation target.

3.2 Tokenizer Training

To address the need for a unified discrete token space for both item interactions and review feedback, we first train a semantic tokenizer based on item text. As shown in Fig. 2, we first encode item text into dense semantic representations with an LLM-based text encoder, and then train an RQ-VAE [29] tokenizer to quantize such representations into multi-level discrete codes. In this way, each item can be represented as a semantic ID sequence that serves as the basis for subsequent review-augmented generative recommendation.

Formally, let x_i denote the textual content associated with item $i \in \mathcal{I}$, such as its title, category, or description. We first employ a pretrained LLM-based text encoder to obtain a dense embedding

$$\mathbf{e}_i = E(x_i), \quad (7)$$

where $E(\cdot)$ denotes the text encoder and $\mathbf{e}_i \in \mathbb{R}^d$ is the semantic embedding of item i . Compared with raw item IDs, such text-informed representations provide richer semantic information and serve as the basis for tokenizer learning.

Based on these item embeddings, we train a residual quantization variational autoencoder (RQ-VAE) to discretize the continuous semantic space into a sequence of code indices. Specifically, given an item embedding $\mathbf{e}_i$, the encoder of RQ-VAE first maps it into a latent representation

$$\mathbf{h}_i = g_{\text{enc}}(\mathbf{e}_i), \quad (8)$$

where $g_{\text{enc}}(\cdot)$ denotes the encoder network. We then apply residual quantization over M codebooks $\{\mathcal{C}^{(1)}, \mathcal{C}^{(2)}, \dots, \mathcal{C}^{(M)}\}$ in a coarse-to-fine manner. At the m -th level, the residual representation is quantized by selecting the nearest codeword from the corresponding codebook:

$$q_i^{(m)} = \arg \min_k \left\| \mathbf{r}_i^{(m-1)} - \mathbf{c}_k^{(m)} \right\|_2^2, \quad (9)$$

where $\mathbf{c}_k^{(m)} \in \mathcal{C}^{(m)}$ denotes the k -th codeword in the m -th codebook, and $\mathbf{r}_i^{(m-1)}$ is the residual to be quantized at level m . The residual is updated as

$$\mathbf{r}_i^{(m)} = \mathbf{r}_i^{(m-1)} - \mathbf{c}_{q_i^{(m)}}^{(m)}, \quad (10)$$

with $\mathbf{r}_i^{(0)} = \mathbf{h}_i$. After M levels of quantization, the item representation is approximated by the sum of selected codewords:

$$\hat{\mathbf{h}}_i = \sum_{m=1}^M \mathbf{c}_{q_i^{(m)}}^{(m)}. \quad (11)$$

Accordingly, each item i is represented as a multi-level discrete semantic ID

$$\mathbf{z}(i) = [q_i^{(1)}, q_i^{(2)}, \dots, q_i^{(M)}], \quad (12)$$

which serves as its tokenized representation in the unified generative space. Intuitively, earlier codebooks capture coarse-grained semantics, while later codebooks progressively refine the representation with finer-grained details. Following standard residual quantization objectives [17], we train the RQ-VAE by minimizing the discrepancy between the latent representation and its

quantized approximation, together with codebook and commitment regularization:

$$\begin{aligned}
 \mathcal{L}_{\text{tok}} = & \underbrace{\left\| \mathbf{h}_i - \hat{\mathbf{h}}_i \right\|_2^2}_{\mathcal{L}_{\text{rec}}} \\
 & + \underbrace{\sum_{m=1}^M \left\| \text{sg} \left[\mathbf{r}_i^{(m-1)} \right] - \mathbf{c}_{q_i^{(m)}}^{(m)} \right\|_2^2}_{\mathcal{L}_{\text{code}}} \\
 & + \underbrace{\beta \sum_{m=1}^M \left\| \mathbf{r}_i^{(m-1)} - \text{sg} \left[\mathbf{c}_{q_i^{(m)}}^{(m)} \right] \right\|_2^2}_{\mathcal{L}_{\text{commit}}}.
 \end{aligned} \tag{13}$$

where $\text{sg}[\cdot]$ denotes the stop-gradient operator and β is the commitment coefficient.

3.3 Review-Augmented User Sequence Modeling

To address the limitation that existing generative recommendation models primarily capture what users selected but not why they selected it, we augment the conventional item-only interaction sequence with review feedback, so that each historical interaction is represented by both its item semantic ID and its corresponding review semantic ID. As illustrated in Fig. 2, instead of organizing user behavior as a pure item trajectory, we interleave item and review tokens in chronological order, allowing the generative recommender to condition sequence modeling on both behavioral outcomes and semantic feedback evidence.

Let $\mathbf{z}(i_t)$ and $\mathbf{z}(r_t)$ denote the semantic IDs of item i_t and review r_t , respectively, obtained from the tokenizer trained in the first stage (Section 3.2). For each user u , we construct the review-augmented sequence as

$$\tilde{\mathcal{S}}_u = [\mathbf{z}(i_1), \mathbf{z}(r_1), \mathbf{z}(i_2), \mathbf{z}(r_2), \dots, \mathbf{z}(i_T), \mathbf{z}(r_T)]. \tag{14}$$

Compared with conventional item-only sequences, this formulation allows the model to observe not only what the user interacted with, but also how the user described or evaluated those interactions.

Based on the review-augmented sequence, we construct a unified textual generation task by serializing item SIDs and review SIDs into autoregressive training instances. Specifically, we generate two types of sequence-to-sequence samples from the same review-augmented interaction history. The first type is *next-item SID prediction*, where the input consists of the historical item-review sequence up to step $t - 1$, and the target is the semantic ID sequence of the next item $\mathbf{z}(i_t)$. The second type is *next-review SID prediction*, where the input additionally includes the target item $\mathbf{z}(i_t)$, and the target is the corresponding review semantic ID sequence $\mathbf{z}(r_t)$. In both cases, the model is trained as a standard generative recommender over textualized semantic IDs, rather than as two separate task-specific predictors.

Formally, let $(\mathbf{x}, \mathbf{y})$ denote a generic training instance constructed from the review-augmented sequence, where $\mathbf{x}$ is the serialized input sequence and $\mathbf{y}$ is the target semantic ID sequence to be generated. Then the GR model is optimized with a unified autoregressive generation objective:

$$\mathcal{L}_{\text{seq}} = - \sum_{(\mathbf{x}, \mathbf{y}) \in \mathcal{D}_{\text{seq}}} \log p_{\theta}(\mathbf{y} \mid \mathbf{x}), \tag{15}$$

where $\mathcal{D}_{\text{seq}}$ denotes the collection of all serialized next-item and next-review training instances. Under this formulation, item prediction and review prediction are not treated as separate optimization objectives, but as two forms of textual sequence generation derived from the same review-augmented user behavior.

This unified sequence modeling strategy provides two benefits. First, it exposes the model to a finer-grained generative sequence that interleaves item interactions and review feedback, thereby enriching user sequence modeling beyond pure item transitions. Second, it allows the GR model to learn the semantic dependency between items and their associated feedback within a shared generative space, which lays the foundation for the subsequent item-centric task alignment stage.

3.4 Item-Centric Task Generation Alignment

To address the risk that review-augmented sequence modeling may blur the task boundary of next-item recommendation and inadvertently drive the model to generate review semantic IDs, we further introduce an item-centric task alignment stage that explicitly aligns the GR model toward target item generation rather than review generation. As illustrated in Fig. 2, the key idea is to treat the target item semantic ID as the *preferred* output and the corresponding review semantic ID as the *rejected* output under the same context, so that review feedback remains informative evidence for item selection instead of becoming a competing generation target.

Formally, let $\mathbf{x}_t$ denote the serialized review-augmented context constructed from the historical sequence before the target interaction, i.e.,

$$\mathbf{x}_t = [\mathbf{z}(i_1), \mathbf{z}(r_1), \dots, \mathbf{z}(i_{t-1}), \mathbf{z}(r_{t-1})]. \quad (16)$$

For each target interaction at step t , we form a preference pair

$$(\mathbf{x}_t, \mathbf{y}_t^+, \mathbf{y}_t^-), \quad (17)$$

where

$$\mathbf{y}_t^+ = \mathbf{z}(i_t), \quad \mathbf{y}_t^- = \mathbf{z}(r_t). \quad (18)$$

Here, $\mathbf{y}_t^+$ is the preferred output because it corresponds to the next target item that should be recommended, while $\mathbf{y}_t^-$ is the rejected output because it corresponds to review feedback, which should inform recommendation but should not replace the recommendation target itself. Based on these pairs, we construct a training set

$$\mathcal{D}_{\text{dpo}} = \{(\mathbf{x}_t, \mathbf{y}_t^+, \mathbf{y}_t^-) \mid u \in \mathcal{U}, 2 \leq t \leq T\}. \quad (19)$$

We then apply DPO [27] to align the GR model toward the preferred item output. Let π_θ denote the current GR model and π_{ref} denote a reference model. To improve readability, we first define the relative preference score between the target item and the review signal under the same context as

$$\Delta_\theta(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) = \log \frac{\pi_\theta(\mathbf{y}^+ \mid \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}^+ \mid \mathbf{x})} - \log \frac{\pi_\theta(\mathbf{y}^- \mid \mathbf{x})}{\pi_{\text{ref}}(\mathbf{y}^- \mid \mathbf{x})}. \quad (20)$$

For each preference triple $(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \in \mathcal{D}_{\text{dpo}}$, the DPO loss is then written as

$$\ell_{\text{DPO}}(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) = -\log \sigma(\beta \Delta_\theta(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-)), \quad (21)$$

and the overall objective is

$$\mathcal{L}_{\text{DPO}} = \mathbb{E}_{(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-) \sim \mathcal{D}_{\text{dpo}}} [\ell_{\text{DPO}}(\mathbf{x}, \mathbf{y}^+, \mathbf{y}^-)]. \quad (22)$$

Here, $\sigma(\cdot)$ denotes the sigmoid function, and β is a temperature coefficient that controls the sharpness of optimization.

This objective explicitly encourages the GR model to assign a higher conditional likelihood to the target item semantic ID than to the review semantic ID under the same review-augmented context.

In this way, review feedback is not removed from the generative sequence; instead, it is retained as contextual evidence while the model is aligned to preserve the item-centric recommendation objective. Consequently, the task alignment stage complements the previous sequence modeling stage: the latter enriches the user sequence with semantic feedback, whereas the former ensures that such feedback improves item generation without shifting the task from *next-item prediction* to *review generation*.

4 Experiment

In this section, we conduct comprehensive experiments to evaluate the proposed **RAGR** framework. The experiments are designed to answer the following five research questions:

- **RQ1:** To what extent can RAGR improve the performance of existing generative recommendation backbones on next-item recommendation?
- **RQ2:** How effective are the two key components of RAGR, namely *Review-Augmented User Sequence Modeling* and *Item-Centric Task Generation Alignment*?
- **RQ3:** Which tokenizer training strategy is most effective for RAGR: training RQ-VAE on item text only, review text only, or both item and review text?
- **RQ4:** How sensitive is RAGR to the number of semantic ID tokens used in the tokenizer?
- **RQ5:** How sensitive is RAGR to the hyperparameter settings of DPO-based task alignment?

We first present the experimental settings, including datasets, baselines, evaluation metrics, and implementation details. We then answer the above research questions through a series of experiments.

4.1 Experimental Settings

Table 1. Statistics of the datasets after preprocessing.

Datasets	#Users	#Items	#Inter.	#RAGR Train	#Val.	#Test
Beauty	22,363	12,101	198,502	285,189	22,363	22,363
Toys	19,412	11,924	167,597	238,134	19,412	19,412
Sport	35,598	18,357	296,337	414,684	35,598	35,598

represents the number of users, items, interactions, and samples.

4.1.1 Datasets. To evaluate the effectiveness of **RAGR**, we conduct experiments on three benchmark datasets from the Amazon review datasets [10]: Amazon-Beauty (Beauty for short), Amazon-Sports and Outdoors (Sport for short), and Amazon-Toys and Games (Toys for short).¹ These datasets contain both user–item interaction records and user-written reviews, making them particularly suitable for evaluating review-augmented generative recommendation. For each dataset, we organize user behaviors into chronological interaction sequences, where each interaction is paired with its corresponding review text. Following common practice in sequential recommendation, we remove users and items with fewer than five interactions to ensure sufficient behavioral context. We then sort all interactions by timestamp and adopt the leave-one-out evaluation protocol [16, 29]: for each user, the last interaction is used for testing, the second last interaction is used for validation, and the remaining interactions are used for training. In our setting, the review text associated with each historical interaction is retained, so that both item interactions and post-decision feedback can be incorporated into the review-augmented user sequence. Table 1 summarizes the statistics of the three datasets.

¹<https://cseweb.ucsd.edu/~jmcauley/datasets/amazon/links.html>

4.1.2 Baselines. To comprehensively evaluate the proposed method, we compare RAGR with two groups of baselines, including representative sequential recommenders and recent generative recommendation models.

Sequential recommenders.

- **GRU4Rec** [11]: GRU4Rec is a classical sequential recommender that models user interaction sequences with gated recurrent units (GRUs). By recurrently updating hidden states along the interaction trajectory, it captures short-term sequential dependency and predicts the next item based on the encoded user state.
- **BERT4Rec** [32]: BERT4Rec extends bidirectional Transformer modeling to sequential recommendation. Instead of relying on unidirectional next-step prediction, it adopts an objective to learn bidirectional contextual representations over user interaction sequences, thereby capturing richer dependency patterns among historical items.
- **SASRec** [16]: SASRec is a self-attention-based sequential recommender that uses Transformer-style attention blocks to model long-range dependencies in user interaction histories. By selectively attending to relevant past items, it effectively captures both short-term and long-term sequential signals for next-item prediction.
- **S³-Rec** [50]: S³-Rec enhances sequential recommendation through self-supervised representation learning. It introduces multiple auxiliary pretraining tasks over item attributes, item co-occurrence, and sequence context, so that the model can learn more informative sequence representations for next-item prediction.

Generative recommenders.

- **TIGER** [29]: TIGER is a representative generative recommender that tokenizes each item into a multi-level semantic ID and reformulates recommendation as an autoregressive generation problem. Instead of predicting over a fixed item vocabulary, it generates the semantic ID sequence of the next target item conditioned on the historical item sequence, thereby bridging generative retrieval and sequential recommendation.
- **LETTER** [36]: LETTER further improves generative recommendation by learning more effective semantic tokenization for items. It enhances the quality of item semantic IDs and the corresponding generative retrieval process, leading to stronger next-item prediction performance under the generative recommendation paradigm.

4.1.3 Evaluation Metrics. Following common practice in sequential recommendation [16, 29], we evaluate all methods with top- K ranking metrics, including **HIT@K** and **NDCG@K**, where $K \in \{5, 10, 20\}$. For each test instance, the model ranks the ground-truth next item together with a set of sampled negative items, and the ranking performance is then measured accordingly.

HIT@K measures whether the ground-truth item appears in the top- K ranked list:

$$\text{HIT@K} = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}(\text{rank}_q \leq K), \quad (23)$$

where Q denotes the set of evaluation instances, rank_q is the rank position of the ground-truth item for instance q , and $\mathbf{1}(\cdot)$ is the indicator function.

NDCG@K further takes the ranking position of the ground-truth item into account and assigns higher scores when the item is ranked closer to the top:

$$\text{NDCG@K} = \frac{1}{|Q|} \sum_{q \in Q} \frac{\mathbf{1}(\text{rank}_q \leq K)}{\log_2(\text{rank}_q + 1)}. \quad (24)$$

4.1.4 Implementation Details. To verify the effectiveness and generality of RAGR across different GR backbones, we instantiate RAGR on two representative GR models, resulting in **TIGER+RAGR** and **LETTER+RAGR**. For a fair comparison, all baselines are trained and evaluated under the same data split and evaluation protocol. For conventional sequential recommenders, only item interaction sequences are used as input, following their original settings. For generative recommenders, we adopt their standard item tokenization and autoregressive prediction settings, while our method further incorporates review feedback into the unified generative sequence.

For the **tokenizer training** stage, we use T5² as a text encoder to obtain embeddings for items and reviews. For the RQ-VAE architecture, we use a six-layer encoder/decoder MLP with hidden dimensions [2048, 1024, 512, 256, 128, 64]. The number of residual codebooks is set to 4, and each codebook contains 256 codewords, and the codeword dimension set to 32. We enable k-means initialization for the codebooks with 100 iterations. The number of clusters is set to 10. In our implementation, the RQ-VAE is optimized with the AdamW optimizer, using a learning rate of 1×10^{-3} , a batch size of 2048, and a weight decay of 1×10^{-4} . The tokenizer is trained for at most 2000 epochs, with evaluation of the collision rate.

For **review-augmented user sequence modeling**, we interleave item semantic IDs and review semantic IDs in chronological order to construct the input sequence. The maximum history length is set to 20, and longer interaction histories are truncated from the left. For the GR backbone, we use T5 [28] as the base generative model. The model is trained for 200 epochs using the AdamW optimizer, with a learning rate of 1×10^{-3} , a per-device batch size of 256, gradient accumulation steps of 2, and a weight decay of 0.01. We adopt a cosine learning rate scheduler with a warmup ratio of 0.01.

For **item-centric task generation alignment**, we initialize the aligned model from the GR model trained in the previous stage, and use it as the policy model in DPO. The β in Eq. (21) is set to 0.5 to 0.7 with sensitivity analysis in Sections 4.6), and the reference model is fixed during alignment training. The model is trained using the AdamW optimizer, with a learning rate of 1×10^{-6} , a per-device batch size of 256.

All experiments were conducted on an Ubuntu server equipped with eight NVIDIA RTX PRO 6000 GPUs (96 GB). For the main performance experiment, all results are the average of three runs with different random seeds: 42, 43, and 44.

4.2 Overall Performance (RQ1)

To answer RQ1, we conducted a comprehensive performance comparison against SR baselines. Table 2 reports the overall performance comparison on Beauty, Toys, and Sport. Several observations can be drawn.

First, RAGR consistently improves both GR backbones, namely TIGER and LETTER, across all datasets and all evaluation metrics. For *TIGER*, the relative improvements range from 7% to 26%. In particular, on Beauty, TIGER+RAGR improves TIGER by 13%/15% on HIT@5/NDCG@5 and by 9%/11% on HIT@20/NDCG@20. On Toys, the gains are even larger, reaching 24% on HIT@5 and 26% on NDCG@5, while remaining consistently above 20% on most metrics. On Sport, although the absolute gains are relatively smaller, RAGR still achieves stable improvements, ranging from 8% to 19%. For *LETTER*, RAGR also brings consistent improvements on all three datasets, with relative gains ranging from 7% to 20%. These results demonstrate that the proposed framework is not tied to a specific GR backbone, but can be effectively generalized across different generative recommendation architectures.

²<https://huggingface.co/sentence-transformers/sentence-t5-base>

Table 2. Performance comparison on Beauty, Toys, and Sport. *Imp.* denotes the relative improvement of RAGR over its corresponding backbone model in each metric.

Dataset	Metric	GRU4Rec	BERT4Rec	SASRec	S ³ -Rec	TIGER +RAGR	TIGER Imp.	LETTER +RAGR	LETTER Imp.
Beauty	HIT@5	0.0174	0.0202	0.0377	0.0361	0.0386	0.0435* 13%	0.0371	0.0446* 20%
	NDCG@5	0.0108	0.0111	0.0220	0.0228	0.0254	0.0292* 15%	0.0253	0.0294* 16%
	HIT@10	0.0323	0.0311	0.0605	0.0598	0.0607	0.0649* 7%	0.0582	0.0677* 16%
	NDCG@10	0.0156	0.0155	0.0303	0.0305	0.0325	0.0361* 11%	0.0321	0.0370* 15%
	HIT@20	0.0513	0.0518	0.0933	0.0940	0.0866	0.0944* 9%	0.0885	0.1019* 15%
	NDCG@20	0.0204	0.0207	0.0385	0.0391	0.0391	0.0435* 11%	0.0397	0.0455* 15%
Toys	HIT@5	0.0147	0.0140	0.0369	0.0376	0.0331	0.0410* 24%	0.0321	0.0386* 20%
	NDCG@5	0.0095	0.0082	0.0217	0.0238	0.0206	0.0259* 26%	0.0210	0.0243* 16%
	HIT@10	0.0257	0.0177	0.0591	0.0604	0.0526	0.0630* 20%	0.0512	0.0604* 18%
	NDCG@10	0.0130	0.0090	0.0289	0.0311	0.0269	0.0329* 22%	0.0272	0.0312* 15%
	HIT@20	0.0420	0.0333	0.0860	0.0910	0.0781	0.0934* 20%	0.0770	0.0902* 17%
	NDCG@20	0.0172	0.0104	0.0356	0.0388	0.0333	0.0406* 22%	0.0337	0.0387* 15%
Sport	HIT@5	0.0152	0.0114	0.0212	0.0204	0.0231	0.0267* 16%	0.0240	0.0264* 10%
	NDCG@5	0.0102	0.0085	0.0138	0.0136	0.0148	0.0176* 19%	0.0156	0.0172* 10%
	HIT@10	0.0231	0.0224	0.0311	0.0306	0.0385	0.0415* 8%	0.0403	0.0433* 7%
	NDCG@10	0.0127	0.0091	0.0177	0.0169	0.0197	0.0224* 14%	0.0209	0.0227* 9%
	HIT@20	0.0359	0.0300	0.0506	0.0483	0.0564	0.0619* 10%	0.0612	0.0661* 8%
	NDCG@20	0.0159	0.0080	0.0220	0.0213	0.0242	0.0274* 13%	0.0262	0.0284* 9%

“*” indicates statistically significant improvements (two-sided t-test with $p < 0.05$) over the corresponding backbone model.

Second, RAGR not only improves strong generative baselines, but also achieves the best overall performance against conventional sequential recommenders. On all three datasets, both TIGER+RAGR and LETTER+RAGR outperform GRU4Rec, BERT4Rec, SASRec, and S³-Rec across all evaluation metrics. This is notable because these SR baselines already represent strong and diverse modeling paradigms, including recurrent, Transformer-based, and self-supervised approaches. The consistent superiority of RAGR therefore suggests that incorporating review feedback into the generative sequence provides additional predictive signals beyond what can be captured by item-only interaction modeling.

Third, review-augmented sequence modeling appears to be especially beneficial in scenarios where textual feedback carries stronger preference signals. This pattern is most evident on Toys, where the gains of RAGR are consistently larger than those on Beauty and Sport. Moreover, the improvements are consistently observed on both HIT@K and NDCG@K, indicating that RAGR improves not only the ability to retrieve the ground-truth next item, but also the quality of its ranking. These results further support the effectiveness of integrating review feedback directly into the generative sequence.

4.3 Ablation Study (RQ2)

To answer RQ2 and understand the contribution of each component in RAGR, we conduct an ablation study on two GR backbones. As illustrated in Fig. 3, we progressively extend the training paradigm from the original item-only generative recommendation to the full RAGR framework. Specifically, the compared variants are defined as follows:

- **+Input:** The original item-only input sequence is augmented with review semantic IDs, while the original next-item prediction target remains unchanged.
- **+Task:** Review prediction is further introduced into the textual generation process, so that both next-item SID prediction and next-review SID prediction are jointly modeled within the same unified sequence generation framework.
- **+RAGR:** The proposed item-centric task alignment is further incorporated, where the target item SID is treated as the chosen output and the corresponding review SID is treated as the rejected output in DPO training.

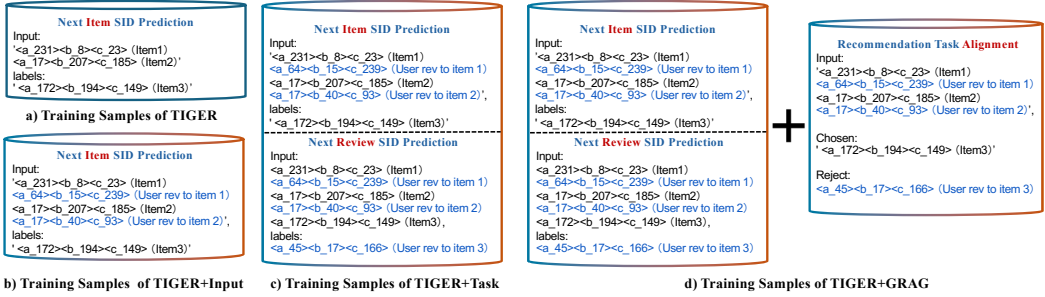


Fig. 3. Illustration of the progressively enhanced training paradigms in the ablation study. Starting from the original item-only training of TIGER, we gradually introduce review-augmented input (**+Input**), review-augmented task modeling (**+Task**), and finally the full RAGR framework with item-centric DPO alignment (**+RAGR**).

The results are reported in Table 3, from which several observations can be made. **First, simply expanding the input sequence with review signals (+Input) does not improve performance; instead, it consistently degrades both TIGER and LETTER on all three datasets.** For example, on Beauty, TIGER drops from 0.0386 to 0.0260 in HIT@5, and LETTER drops from 0.0371 to 0.0311. Similar degradation can also be observed on Toys and Sport, where the +Input variant underperforms the original backbone across all evaluation metrics. This result indicates that review information cannot be effectively utilized by merely inserting review semantic IDs into the input side while keeping the original next-item prediction objective unchanged. In this setting, review tokens introduce additional heterogeneous signals into the user sequence, but the model is not explicitly trained to understand their role or predict them within the generative process. As a result, the augmented sequence may disturb the original item-transition patterns and increase the difficulty of sequence modeling. This observation confirms that review augmentation is not a trivial input-level extension: without a matched training objective, review SIDs may behave more like noise than useful preference evidence.

Second, after extending the training objective to include review prediction (+Task), performance improves substantially over both the original backbone and the +Input variant. For instance, on Toys, TIGER+Task reaches 0.0405/0.0258 on HIT@5/NDCCG@5, clearly outperforming both TIGER and TIGER+Input; similarly, LETTER+Task also achieves large gains over the corresponding backbone. The same trend is consistently observed on Beauty and Sport, suggesting that the benefit of review modeling is robust across datasets and backbones. Compared with +Input, the +Task variant does not merely expose the model to review SIDs, but further requires the model to generate them as part of the unified sequence modeling process. This makes review tokens meaningful training targets rather than passive contextual tokens, encouraging the model to learn the semantic correspondence between user interactions and their associated

Table 3. Ablation study of input expansion, task expansion, and task alignment on Beauty, Toys, and Sport.

Dataset	Method	H@5	N@5	H@10	N@10	H@20	N@20
Beauty	TIGER	0.0386	0.0254	0.0607	0.0325	0.0866	0.0391
	TIGER+Input	0.0260	0.0164	0.0417	0.0215	0.0619	0.0265
	TIGER+Task	0.0427	0.0289	0.0649	0.0361	0.0938	0.0434
	TIGER+RAGR	0.0435	0.0292	0.0649	0.0361	0.0944	0.0435
	LETTER	0.0371	0.0253	0.0582	0.0321	0.0885	0.0397
	LETTER+Input	0.0311	0.0205	0.0511	0.0269	0.0778	0.0336
	LETTER+Task	0.0421	0.0280	0.0634	0.0353	0.0971	0.0435
	LETTER+RAGR	0.0446	0.0294	0.0677	0.0370	0.1019	0.0455
Toys	TIGER	0.0331	0.0206	0.0526	0.0269	0.0781	0.0333
	TIGER+Input	0.0273	0.0174	0.0424	0.0223	0.0676	0.0287
	TIGER+Task	0.0405	0.0258	0.0624	0.0327	0.0930	0.0402
	TIGER+RAGR	0.0410	0.0259	0.0630	0.0329	0.0934	0.0406
	LETTER	0.0321	0.0210	0.0512	0.0272	0.0770	0.0337
	LETTER+Input	0.0273	0.0174	0.0424	0.0223	0.0676	0.0287
	LETTER+Task	0.0382	0.0240	0.0600	0.0310	0.0901	0.0386
	LETTER+RAGR	0.0386	0.0243	0.0604	0.0312	0.0902	0.0387
Sport	TIGER	0.0231	0.0148	0.0385	0.0197	0.0564	0.0242
	TIGER+Input	0.0191	0.0120	0.0317	0.0161	0.0482	0.0202
	TIGER+Task	0.0266	0.0170	0.0413	0.0223	0.0606	0.0271
	TIGER+RAGR	0.0267	0.0176	0.0415	0.0224	0.0619	0.0274
	LETTER	0.0240	0.0156	0.0403	0.0209	0.0612	0.0262
	LETTER+Input	0.0194	0.0126	0.0310	0.0164	0.0484	0.0208
	LETTER+Task	0.0263	0.0172	0.0430	0.0226	0.0658	0.0284
	LETTER+RAGR	0.0264	0.0172	0.0433	0.0227	0.0661	0.0284

feedback. These results indicate that review SIDs become beneficial only when they are explicitly incorporated into the generative task, allowing the model to jointly model item transitions and review-based preference signals within the same token space.

Third, adding the proposed item-centric task alignment further improves performance in most settings and consistently yields the best overall results. On Beauty, TIGER+RAGR improves over TIGER+Task from 0.0427 to 0.0435 in HIT@5 and from 0.0434 to 0.0435 in NDCG@20, while LETTER+RAGR further improves over LETTER+Task from 0.0421 to 0.0446 in HIT@5 and from 0.0971 to 0.1019 in HIT@20. On Toys, the improvement from +Task to +RAGR is also consistent for both backbones, and on Sport, although the gains are relatively smaller, RAGR still maintains stable advantages over +Task in most metrics. These results show that task expansion alone is not sufficient to fully address the tension introduced by mixed item-review generation. Once review prediction is added, the model may allocate part of its generation capacity to review SIDs, which are useful as preference evidence but should not dominate the next-item recommendation objective. By introducing DPO-based item-centric alignment, RAGR explicitly encourages the model to prefer target item SIDs over review SIDs at prediction positions. Therefore, review tokens are used to enrich the generative context while the final optimization direction remains anchored to item

recommendation. This verifies that the proposed alignment mechanism is necessary for turning review prediction into an effective auxiliary signal rather than a competing generation objective.

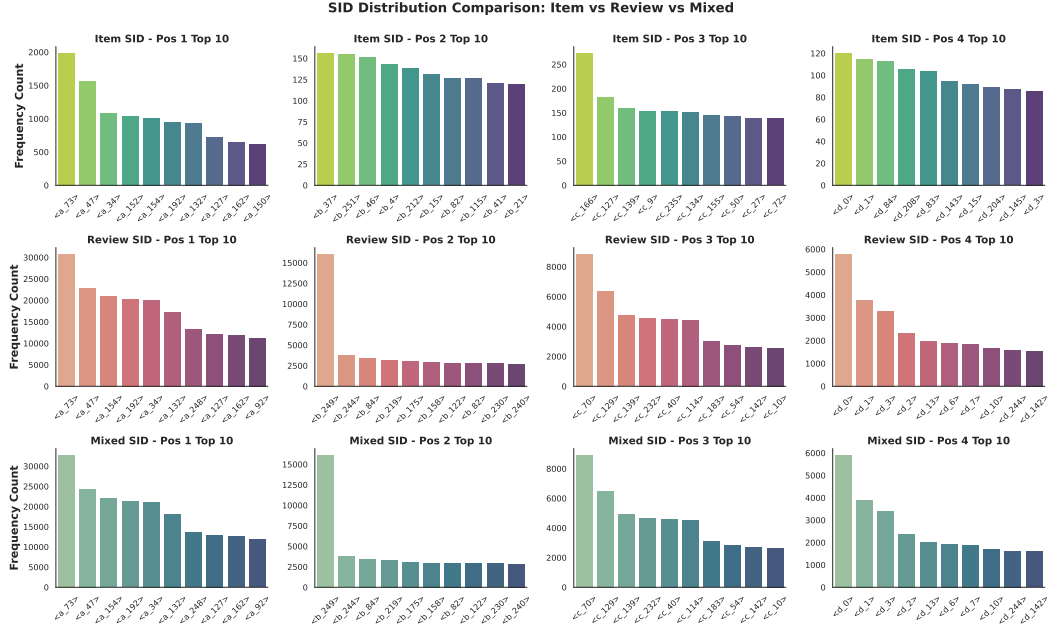


Fig. 4. Comparison of Top-10 SID frequency distributions across item-only, review-only, and mixed item-review sequences.

To further understand why review augmentation improves recommendation performance, we compare the Top-10 SID frequency distributions of item-only, review-only, and mixed item-review sequences across four token positions, as shown in Fig. 4. Several observations can be made. First, item SIDs and review SIDs exhibit markedly different frequency patterns. The item-side distribution is relatively concentrated around a limited number of frequently occurring SIDs, especially at the first and third positions, indicating that item tokens mainly reflect recurring semantic patterns in product descriptions. In contrast, review SIDs show a much heavier concentration on several high-frequency tokens, particularly at the first position, while still introducing a different set of semantic IDs from the item side. This suggests that reviews are not merely redundant textual descriptions of items; instead, they encode user-side evaluative semantics that are distributed differently from item content.

Second, the mixed item-review sequence inherits signals from both sides. Compared with the item-only SID distribution, the mixed distribution activates additional review-related SIDs and reshapes the frequency profile across token positions. This indicates that incorporating reviews does not simply enlarge the training sequence length, but changes the semantic composition of the sequence itself. As a result, the model is exposed to a richer token space that contains both behavioral outcomes, represented by item SIDs, and user preference rationales, represented by review SIDs.

Third, this distributional difference helps explain the effectiveness of RAGR. Existing SID-based GR methods mainly improve the expressiveness of item representation, but the user sequence remains an item-only trajectory. By contrast, review augmentation expands the sequence from a

purely behavioral sequence to a mixed behavioral-semantic sequence. This allows the autoregressive model to learn from semantic feedback that directly reflects users' post-interaction evaluations, such as satisfaction, dissatisfaction, usage experience, and fine-grained preferences. Therefore, the gains of **+Task** and **+RAGR** in Table 3 can be attributed not only to the additional textual information introduced by reviews, but also to the improved utilization of the unified SID space during training.

4.4 Tokenizer Experiment (RQ3)

To answer RQ3, we consider three tokenizer training strategies, as illustrated in Fig. 5. Since the proposed framework relies on a unified semantic ID space for both item interactions and review feedback, the quality of the tokenizer is critical to the effectiveness of subsequent review-augmented sequence modeling and item-centric task alignment.

- **Item Text-Only:** the RQ-VAE tokenizer is trained only on item text embeddings.
- **Review Text-Only:** the RQ-VAE tokenizer is trained only on review text embeddings.
- **Item and Review Text:** the RQ-VAE tokenizer is trained on the combination of item text and review text embeddings.

To provide a more direct comparison, Table 4 summarizes the training efficiency and collision rates of the three tokenizer training strategies across the three datasets.

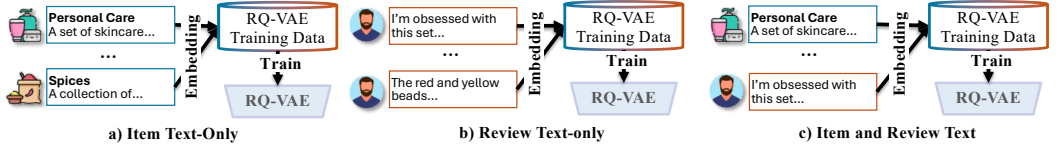


Fig. 5. Illustration of the three tokenizer: training RQ-VAE on item text only, review text only, and both item and review text.

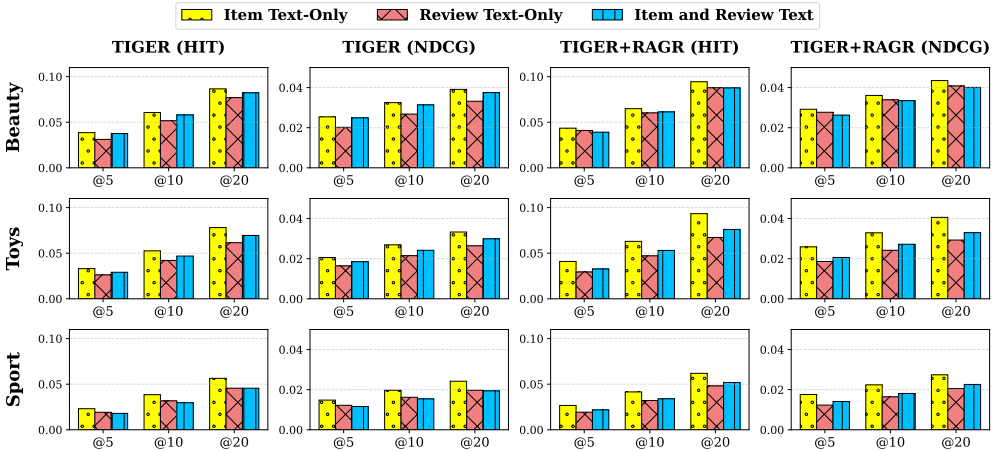


Fig. 6. Impact of different tokenizer training strategies on downstream recommendation performance. We compare tokenizers trained on *Item Text-Only*, *Review Text-Only*, and *Item and Review Text* under both TIGER and TIGER+RAGR.

Table 4. Comparison of different tokenizer training strategies in terms of training efficiency and collision rate (Col.).

Tokenizer	Training Time	Beauty Col.	Toys Col.	Sport Col.
Item Text-Only	2s/epoch	0.0009	0.0011	0.0025
Review Text-Only	29s/epoch	0.0009	0.0012	0.0024
Item and Review Text	33s/epoch	0.0008	0.0044	0.0005

Overall, the tokenizer experiment leads to three main observations. First, **Item Text-Only** consistently achieves the strongest downstream recommendation performance in most settings, for both the original TIGER backbone and the review-augmented TIGER+RAGR variant. As shown in Fig. 6, the item-text tokenizer yields the highest HIT and NDCG values on Beauty, Toys, and Sport in the majority of cases. This suggests that, for generative recommendation, learning semantic IDs from item text provides a more stable and effective token space for next-item prediction.

Second, **Review Text-Only** generally performs the worst among the three strategies. Although the review text contains rich user preference signals, using it alone to train the tokenizer appears to shift the item sid distribution, which weakens the quality of item representations and harms downstream recommendation performance. In contrast, **Item and Review Text** often performs better than **Review Text-Only**, but still does not consistently surpass **Item Text-Only**. This indicates that directly mixing review text into tokenizer training does not necessarily lead to a better semantic ID space, even though review feedback is beneficial at the sequence modeling stage.

Third, Table 4 shows that **Item Text-Only** is also substantially more efficient, requiring only 2s per epoch, compared with 29s and 33s for **Review Text-Only** and **Item and Review Text**, respectively. Although the collision rates do not always perfectly align with downstream recommendation performance, the item-text tokenizer provides the best overall trade-off between efficiency and effectiveness. Therefore, in the remaining experiments, we adopt **Item Text-Only** as the default tokenizer training strategy.

4.5 SID Sensitivity Experiment (RQ4)

Table 5. Collision rates under different SID numbers on the three datasets.

SID Num	Beauty Col.	Toys Col.	Sport Col.
3	0.0121	0.0017	0.0045
4	0.0009	0.0012	0.0026
5	0.0007	0.0003	0.0018

To answer **RQ4**, we investigate how the number of semantic ID tokens affects the effectiveness of RAGR. Intuitively, the number of semantic ID tokens controls the granularity of the discrete semantic space. Fewer tokens lead to a smaller semantic space and thus a higher risk of collisions, whereas more tokens improve semantic distinctiveness but may increase the difficulty of autoregressive generation. Therefore, studying the sensitivity of RAGR to the SID number is important for balancing representation quality and recommendation performance.

We vary the SID number from 3 to 5 and report both the tokenizer collision rate and the downstream recommendation performance. Table 5 summarizes the collision rate under different SID numbers on Beauty, Toys, and Sport. As expected, increasing the SID number consistently

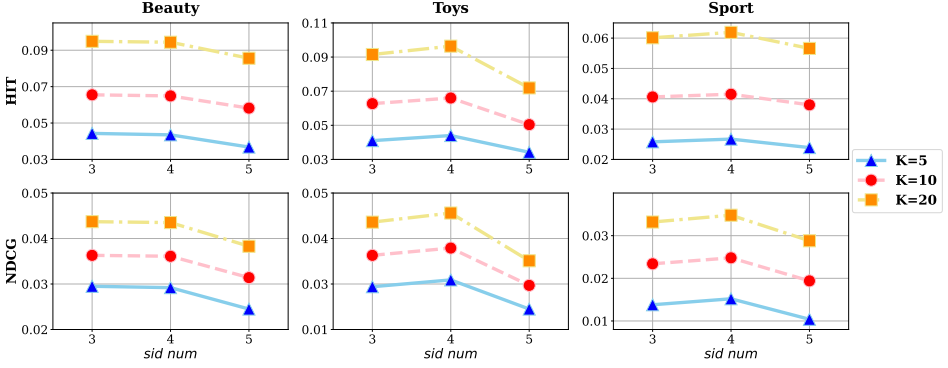


Fig. 7. Impact of the number of semantic ID tokens on recommendation performance. We report HIT@K and NDCG@K ($K \in \{5, 10, 20\}$) on Beauty, Toys, and Sport under different SID numbers.

reduces the collision rate across all datasets. For example, on Beauty, the collision rate decreases from 0.0121 when using 3 semantic ID tokens to 0.0009 and 0.0007 when using 4 and 5 tokens, respectively. Similar trends are observed on Toys and Sport. This confirms that longer semantic IDs indeed improve token distinctiveness and reduce representation collisions. However, a lower collision rate does not necessarily translate into better recommendation performance. Figure 7 reports the downstream results, from which several observations can be drawn.

First, using 4 semantic ID tokens achieves the best overall performance across the three datasets and both metrics. On Beauty, Toys, and Sport, SID number 4 consistently gives the strongest HIT and NDCG results, indicating that it provides the most effective balance between semantic distinctiveness and generation difficulty. Second, the relationship between SID length and performance is clearly non-monotonic. When the SID number is too small, the collision rate increases, leading to multiple items sharing the same or highly similar semantic IDs. In this case, the tokenizer loses item-level discriminability, and the model can only learn a coarse semantic partition of the item space. For example, on Beauty, SID number 3 still yields relatively competitive results, even though its collision rate is much higher than that of SID number 4. A plausible explanation is that the higher collision rate causes semantically similar or high-frequency items to be merged into shared semantic IDs, which introduces a smoothing effect and partially alleviates data sparsity in generation. However, such gains come at the cost of reduced item distinguishability, which limits further performance improvement. Third, increasing the SID number from 4 to 5 further reduces collision rates, but consistently hurts recommendation performance. This suggests that once semantic collisions are sufficiently controlled, making the semantic ID sequence even longer mainly enlarges the generation space and increases autoregressive decoding difficulty. As a result, the model faces a harder generation problem without receiving commensurate gains in representation quality.

Overall, the results suggest that SID length controls a trade-off between semantic uniqueness and generation complexity. Too few tokens lead to excessive collisions and insufficient item discrimination, while too many tokens make next-item generation unnecessarily sparse and difficult. Therefore, a moderate SID length is most desirable in practice. Based on these results, we set the SID number to 4 as the default configuration of RAGR.

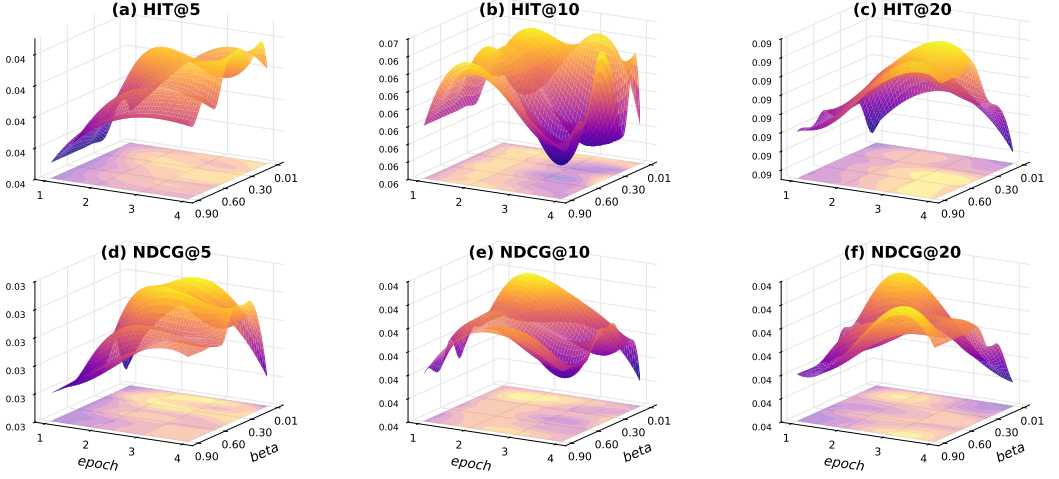


Fig. 8. Sensitivity analysis of the DPO-based task alignment with respect to the preference coefficient β and training epoch. We report HIT@{5, 10, 20} and NDCG@{5, 10, 20} under different hyperparameter combinations.

4.6 DPO Sensitivity Experiment (RQ5)

To answer **RQ5**, we investigate how sensitive RAGR is to the hyperparameter settings of DPO-based task alignment. In particular, we focus on the preference coefficient β , which controls the sharpness of preference optimization, and the training epoch, which determines the extent to which the GR backbone is further aligned toward item-centric generation. Since DPO is introduced to preserve the task boundary of *next-item recommendation* after review signals are incorporated, understanding its hyperparameter sensitivity is important for assessing the stability and robustness of the proposed alignment strategy. Figure 8 visualizes the performance landscape under different combinations of β and training epoch in terms of HIT@{5, 10, 20} and NDCG@{5, 10, 20}. Several observations can be made.

First, the performance surface is generally smooth rather than highly irregular, indicating that the proposed DPO-based alignment is reasonably stable over a broad range of hyperparameter settings. Second, for most evaluation metrics, moderate values of β consistently lead to better performance than overly small or overly large values. This suggests that if β is too small, the preference signal becomes too weak to effectively enforce item-centric alignment; in contrast, if β is too large, the model may overemphasize the chosen-rejected contrast and thus harm the original generative recommendation capability. Third, the interaction between β and training epoch further reveals that the best performance is usually achieved within a moderate training range, rather than at the earliest or latest alignment epochs. This indicates that DPO-based task alignment should be sufficiently strong to shift the model toward target item generation, but not so strong as to overfit the preference pairs. In other words, the sensitivity analysis confirms that the effectiveness of RAGR does not rely on an extremely narrow hyperparameter region, while also showing that proper tuning of β and training epoch is beneficial for obtaining the best performance.

Overall, the results demonstrate that the proposed item-centric task alignment is robust to hyperparameter variation and can stably improve recommendation performance across a reasonably wide range of settings. Based on the validation results, we select the corresponding β and epoch

combination that achieves the best trade-off between performance and stability as the default configuration in our experiments.

5 Related work

5.1 Sequential Recommendation

Sequential recommendation (SR) aims to model users' evolving preferences from their historical interaction sequences and predict the next item they are likely to engage with. As one of the most fundamental paradigms in recommender systems [25, 33], it has been extensively studied under the *next-item prediction* formulation, which explicitly reflects the dynamic nature of user interests and has become a standard testbed for modeling sequential behavior.

Existing sequential recommendation methods mainly differ in how they encode the interaction sequence and capture dependency patterns within it. Early studies relied on recurrent architectures, such as GRU4Rec [11], to model sequential transitions through hidden-state propagation, establishing the foundation for sequence-aware recommendation. Later, Transformer-based models became dominant, including SASRec [16] and BERT4Rec [32], which leverage self-attention and bidirectional contextual modeling to learn more expressive sequence representations and capture long-range dependency more effectively. Building on these backbones, subsequent work further improved robustness and generalization through self-supervised learning and contrastive objectives. Representative examples include S³-Rec [50], which introduces auxiliary pretext tasks to strengthen sequence representation learning, and contrastive learning approaches [42], which encourage more discriminative user modeling under augmented views of behavior sequences.

Beyond architecture design, recent studies have also explored richer inductive biases for sequential recommendation. Temporal modeling methods [44] incorporate time intervals and temporal contexts into user modeling, dynamic graph approaches [45] capture high-order collaborative dependencies beyond linear sequences, and hyper adapter mechanisms [19] improve parameter adaptation under heterogeneous recommendation scenarios. More recently, LLM-enhanced representations [21, 34] have been introduced to inject stronger semantic priors into recommendation models, further broadening the representational capacity of SR. Taken together, these advances have substantially improved the ability of SR models to encode user interaction histories and forecast future behaviors.

Despite these advances, most existing SR methods still model user behavior as an item-only interaction sequence and optimize for next-item prediction. **As a result, they mainly capture what users selected, while overlooking richer behavior signals such as reviews and feedback that may help explain why users selected or rejected items.** This limitation is nontrivial, because in real-world recommendation scenarios, user decision-making is rarely reflected by item transitions alone. Users often browse, compare, purchase, and then leave explicit feedback, and such feedback may reveal fine-grained preference cues that are difficult to infer from item co-occurrence patterns alone. In contrast, our work moves beyond item-only sequence modeling by incorporating review feedback into the generative user sequence.

5.2 Generative Recommendation

Generative recommendation (GR) reformulates recommendation as a generation problem, where the model produces the target item through autoregressive decoding rather than ranking over a fixed item vocabulary. By replacing conventional scoring-and-ranking pipelines with sequence generation, GR opens up a new modeling perspective for recommendation and has recently emerged as a rapidly growing research direction. Recent surveys have highlighted that progress in this

paradigm is mainly driven by three closely related aspects: tokenization, generative architecture, and optimization [4, 14, 18].

Existing GR methods mainly focus on how to represent items as generatable tokens and how to improve the efficiency and effectiveness of generation. A representative line of work learns *semantic IDs* for items and performs next-item generation over these discrete identifiers, such as TIGER [29] and LETTER [36]. The central idea behind this line is to transform item recommendation into a token generation task by mapping items into structured discrete codes, thereby avoiding direct classification over an extremely large item vocabulary. In this way, GR provides a scalable alternative to conventional recommendation formulations while also creating a unified interface between recommendation and sequence generation.

Building on this idea, more recent studies have further investigated tokenization and decoding strategies for generative recommendation. Representative efforts include ID tokenization for LLM-based recommendation [15, 26], parallel decoding of long semantic IDs [12], contextual tokenization of action sequences [13], unified sparse-dense generative recommendation [43], acceleration-oriented decoding architectures [39], and combine semantic and collaborative code [41]. Collectively, these studies have substantially advanced the tokenization mechanism, decoding efficiency, and architectural design of generative recommendation, making GR increasingly practical and expressive as a recommendation paradigm.

Despite these advances, however, existing GR methods still largely remain within the traditional next-item prediction paradigm. **That is, the user sequence is still predominantly modeled as an item-only interaction sequence, and the generation target remains the next item.** In other words, although GR replaces raw item IDs with semantic IDs and casts recommendation into an autoregressive generation process, it mainly changes *how* items are represented and generated, rather than *what* constitutes the user sequence itself. As a result, the expressive capacity of item representation has been

5.3 Review-Aware Recommendation

Review-aware recommendation aims to leverage user-generated reviews to enrich preference modeling beyond pure interaction signals. As discussed in recent surveys [9, 40], reviews have been widely recognized as a valuable source of semantic evidence for recommendation, because they contain rich information about user opinions, product attributes, and decision rationale.

A large body of prior work incorporates reviews as auxiliary textual features for matching or rating prediction. Early methods jointly modeled users and items with review text to improve recommendation quality [47], while subsequent studies explored aspect-aware modeling [3, 24], review-level explanation and interpretability [2, 5], review property modeling [38], and graph-based or hypergraph-based semantic interaction modeling [20, 31]. At the same time, several studies have critically examined the actual utility of reviews in recommendation and discussed their limitations and possible improvements [30]. More recently, with the rise of large language models, another line of work has attempted to enhance recommendation by exploiting the text understanding and reasoning capability of LLMs, such as recommendation-as-language-processing [7], prompt-based personalized recommendation [22], rationale-enhanced LLM recommendation [37], and review-analysis-based feature enhancement [1].

Despite their differences, most existing review-aware recommendation methods still treat reviews as auxiliary features and side information outside the core recommendation sequence. **In other words, reviews are typically used to enhance user/item representations or improve matching, rather than being incorporated into a unified generative sequence together with item interactions.** In contrast, our work explicitly maps review feedback into the same semantic ID space as items, integrates reviews into the generative user sequence itself, and further

introduces item-centric task alignment to ensure that review signals enhance next-item generation without becoming competing generation targets.

6 Conclusion

In this paper, we take a first step toward rethinking generative recommendation beyond the conventional item-only paradigm. We argue that the longstanding assumption of modeling user sequences solely as item trajectories constitutes a fundamental limitation, as it captures what users choose but overlooks why they choose it. To our knowledge, this work is the first to bring review feedback directly into the generative recommendation paradigm by encoding reviews as tokens within the user sequence itself. Based on this idea, we propose **Review-Augmented Generative Recommendation (RAGR)**, a general framework that augments item-only behavioral sequences with review semantics while preserving the next-item recommendation objective.

Specifically, RAGR first learns a unified semantic ID space for both items and reviews through a shared tokenizer, enabling heterogeneous behavioral and textual signals to be represented in a compatible generative form. It then constructs review-augmented user sequences by chronologically interleaving item semantic IDs with their corresponding review semantic IDs, allowing review feedback to participate directly in autoregressive sequence modeling. To prevent review tokens from becoming competing prediction targets, we further introduce an item-centric task generation alignment strategy based on direct preference optimization, which encourages the model to leverage reviews as contextual evidence while remaining focused on next-item generation.

Extensive experiments on three real-world datasets demonstrate the effectiveness, robustness, and generality of RAGR. Across multiple generative recommendation backbones, RAGR consistently improves recommendation performance over strong item-only baselines, showing that review feedback provides complementary preference information that cannot be fully captured by interaction sequences alone. Moreover, ablation studies verify the necessity of both review-augmented sequence modeling and item-centric alignment, while further analyses on tokenizer choices and alignment hyperparameters suggest that the proposed framework is broadly applicable and stable under different settings.

Overall, this work highlights a promising direction for enriching generative recommendation with more expressive user-side signals. By moving reviews from external auxiliary features into the generative sequence itself, RAGR opens up a new perspective on how behavioral outcomes and preference rationales can be jointly modeled within a unified recommendation framework. In the future, we hope this idea can inspire broader exploration of generative recommendation with richer forms of user feedback, such as search queries, comments, conversations, and multimodal interactions, as well as more fine-grained alignment strategies that further balance semantic understanding and recommendation accuracy.

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